

HR ANALYTICS – PREDICT EMPLOYEE ATTRITION

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INTRODUCTION

Employee attrition is an important issue for organizations because employees leaving their jobs can affect productivity, workload, recruitment, and overall business performance. HR Analytics helps organizations use employee data to understand workforce patterns and support better employee retention planning.

This project focuses on analysing employee data to identify patterns related to employee attrition and developing a machine learning model to predict employee attrition. The project combines Python-based data analysis, Machine Learning, and Power BI visualization to create an end-to-end HR analytics solution.

The project uses the IBM HR Analytics Employee Attrition dataset, which contains **1,470 employee records and 35 columns**. The analysis considers factors such as department, job role, overtime, age, job satisfaction, business travel, income, and years of experience.

ABSTRACT

The **HR Analytics – Predict Employee Attrition** project was developed to analyse employee attrition patterns and predict whether an employee may leave an organization. The project begins with data preprocessing and exploratory data analysis using Python. Different employee-related factors are analysed to understand their relationship with historical attrition patterns.

A **Logistic Regression** machine learning algorithm is used for binary classification because the target variable, Attrition, has two outcomes: Yes and No. The dataset is divided into training and testing data, and the model is evaluated using accuracy, precision, recall, F1 score, and a confusion matrix.

The Logistic Regression model achieved **86.05% accuracy, 61.54% precision, 34.04% recall, and 43.84% F1 score** on the test dataset. A Power BI dashboard was also developed to present the main HR insights using KPI cards and charts.

The completed project demonstrates how data science and business intelligence can be combined to analyse employee attrition and support further HR investigation and

retention planning.

TOOLS USED

Python: Used for data preprocessing, exploratory data analysis, machine learning, and result generation.

Pandas: Used for loading, cleaning, and analysing the employee dataset.

NumPy: Used for numerical operations and data processing.

Matplotlib and Seaborn: Used to create charts and visualize employee attrition patterns.

Scikit-learn: Used for data preprocessing, Logistic Regression, model training, and model evaluation.

Joblib: Used for saving the trained machine learning model and scaler.

Power BI: Used to create the interactive HR Employee Attrition Analytics dashboard.

STEPS INVOLVED IN BUILDING THE PROJECT

1. Data Collection:

The IBM HR Analytics Employee Attrition dataset was selected for the project. It contains employee information and the Attrition target variable.

2. Data Preprocessing:

The dataset was loaded using Python and checked for missing values and duplicate records. Categorical variables were converted into numerical features using one-hot encoding.

3. Feature Scaling:

Numerical features were standardized so that the machine learning model could process the features effectively.

4. Exploratory Data Analysis:

Employee attrition was analysed according to department, job role, overtime, age group, job satisfaction, business travel, years at company, and other employee characteristics.

5. Model Building:

Logistic Regression was selected as the main machine learning algorithm. The dataset was divided into training and testing sets using an 80:20 stratified split.

6. Model Evaluation:

The trained model was evaluated using accuracy, precision, recall, F1 score, and a

confusion matrix.

7. Model Results:

The model achieved **86.05% accuracy**, **61.54% precision**, **34.04% recall**, and **43.84% F1 score**. The confusion matrix contained 237 true negatives, 10 false positives, 31 false negatives, and 16 true positives.

8. Power BI Dashboard:

A professional dashboard named **HR EMPLOYEE ATTRITION ANALYTICS** was created. It includes KPI cards for Total Employees, Employees Left, Attrition Rate, and Employees Working Overtime.

9. Dashboard Visualizations:

Charts were created for Attrition by Department, Attrition by Overtime, Attrition by Job Role, Attrition by Age Group, Attrition by Job Satisfaction, and Attrition by Business Travel.

10. Insight Generation:

The dashboard and analysis were used to identify employee groups and factors showing different historical attrition patterns. These insights can support further investigation by HR teams.

CONCLUSION

The **HR Analytics – Predict Employee Attrition** project successfully demonstrates an end-to-end data science workflow using employee data. Python was used for data preprocessing, exploratory analysis, and machine learning, while Logistic Regression was used to predict employee attrition.

The model achieved **86.05% accuracy** on the test dataset, with precision of **61.54%**, recall of **34.04%**, and F1 score of **43.84%**. The Power BI dashboard provides a clear visual representation of employee attrition patterns across different HR factors.

The project shows how machine learning and data visualization can be used together to understand workforce patterns and support employee retention planning. The model should be treated as a decision-support tool and should be combined with HR knowledge and organizational context before making real-world decisions.